

Open-System / Monitored-System Response Theory for a Class-A Adaptive Fermion Circuit

Internal theory and methods note¹

¹*Class-A fermionic adaptive-circuit project*

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This note consolidates the response-theory discussion surrounding chirality at the monitored class-A domain wall. Its first conclusion is economical: a local number-phase impulse followed by a wall-resolved density measurement is already sufficient to test directed propagation. Fourier transformation along the wall and in monitored circuit time should reveal a low-energy ridge near $\omega = v_w k_y$, with the sign of v_w reversing between the two oppositely oriented walls. This diagnoses chirality but is not, by itself, a quantized Hall coefficient. Its second conclusion is constructive and numerical. For perfect correction the Born-mean covariance obeys an exact affine replacement map, so a time-dependent-flux pump can first be propagated deterministically, with no outcome Monte Carlo, score estimator, or Keldysh calculation. Physical Born trajectories are then sampled only to measure record-to-record fluctuations. Event transfers and regional controller weights define an inexpensive source-subtracted monitored-channel pump. Because the reduced circuit exchanges charge with feedback ancillas, a conventional electrical Hall conductance requires one additional microscopic specification: compile each local controller measurement into charge-conserving Givens gates, a local system-ancilla reset, and the inverse gates, then count charge crossing a cut gate by gate. This produces a direct no-postselection numerical experiment in the genuinely space-time-disordered circuit. Generating-functional language is retained only as optional analytic context. Finally, we record the current implementation status: the exact tangent formulas are already derived in the manuscript appendix, whereas the present H2 production runner still uses batched central $\pm\epsilon$ replays and should regard an analytic tangent as a future upgrade.

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I. PURPOSE, SCOPE, AND BOTTOM LINE

The original question was how a record-wise pattern can prove chirality, and what a density-response theory means in a monitored trajectory ensemble. The discussion subsequently touched five related but inequivalent constructions: individual trajectory response, trajectory-averaged response, Born-ensemble response, density spectroscopy, and flux

insertion. Parts of the answer were already distributed through the current manuscript and campaign documents, but the complete logical chain was not available in one place. This note is that source of truth.

The minimum defensible program is:

1. draw a physical stationary record ξ from the unperturbed Born process;
2. apply an infinitesimal, charge-preserving local number-phase impulse at a source on one wall;
3. propagate the exact covariance tangent along the same ordered record;
4. retain both the conditional density derivative and the chronological derivative of the log Born probability;
5. construct the fixed-record trajectory average and the full Born response separately;
6. window the two walls separately and Fourier transform in the coordinate parallel to the wall and in physical circuit time;
7. require an oriented low-energy ridge, opposite ridge slopes on opposite walls, wall localization, and a bulk or matched-trivial null result.

No current operator is required for this claim. A current-current response, counting field, or adiabatic pump becomes necessary only for a stronger transport or quantization statement. The density susceptibility is a direct chirality diagnostic because a one-dimensional chiral density mode has a single signed dispersion at low energy.

A. Where the material already lives

Table I separates the compact manuscript statements, the campaign authority, the current executable implementation, and the present consolidation. The older file `00_WORKSPACE/CURRENT/prxq_draft/numerical_campaign.tex` contains superseded H2/H3 sample counts and is not used as the operational authority.

II. THE OBJECTS THAT MUST NOT BE CONFLATED

A. Quantum instrument, record, and conditioned state

Let a complete ordered record be

$$\xi = (e_1, m_1; e_2, m_2; \dots; e_D, m_D), \quad (1)$$

TABLE I. Repository map. Paths are relative to the current-workspace root. “Formal” means an exact derivation; “executable” means that the current production source actually evaluates the stated estimator.

Location	Role	Status
<code>prxq_draft/main.tex</code> , Sec. “Chiral dynamics”	Compact paper narrative for modular drift, full Born density response, and fixed-record twist flow.	Formal, condensed
<code>prxq_draft/appendices.tex</code>	Microscopic instrument, exact normalized tangent, chronological score, Born-response identity, fixed-record holonomy, and ensemble distinctions.	Formal
<code>experiment_review/numerical_campaign_legacy_working.tex</code>	Audited H1–H3 run matrix, controls, acceptance gates, and legacy-evidence classification.	Campaign authority
<code>final_production_ready_figure_scripts/_shared_src/fused_chirality_observables.py</code>	H2 live-parent implementation; central differences the density and log likelihood under common frozen records.	Executable, finite difference
<code>final_production_ready_figure_scripts/03_chirality_replay/</code>	H3 representative frozen-record twist circle.	Executable
<code>PaperCampaignSummaries/mean_channel_lindblad_cpu_v2_hybrid_response/</code>	Deterministic Gaussian mean-channel/Lindblad response with no Born records.	Completed special case
This directory	Detailed synthesis, literature comparison, and exact-tangent upgrade specification.	Canonical methods note

where e_t includes the selected controller orbital, target occupation, schedule information, and any exogenous random choice, while m_t is the intrinsic Born outcome. The unnormalized branch map is

$$\mathcal{K}_{\xi,u}(\rho_0) = K_{D,u} \cdots K_{1,u} \rho_0 K_{1,u}^\dagger \cdots K_{D,u}^\dagger. \quad (2)$$

For perturbation parameter u , its trace and normalized state are

$$p_u(\xi) = \text{Tr} \mathcal{K}_{\xi,u}(\rho_0), \quad (3)$$

$$\rho_{u,\xi} = \frac{\mathcal{K}_{\xi,u}(\rho_0)}{p_u(\xi)}. \quad (4)$$

Equation (3) is the many-body Born probability of the entire record. A Born-sampled Monte Carlo ensemble already draws each ξ with this probability. Its trajectories therefore enter ordinary sample means with equal numerical weight; multiplying each sample by its stored $p(\xi)$ a second time would double-weight the ensemble.

Quantum-trajectory unravelings and their equivalence to master-equation averages for linear observables are standard [1, 2]. What matters here is that the adaptive record also changes the future normalized state and hence all later conditional probabilities.

B. Three meanings of response

Let

$$O_{u,\xi} = \text{Tr}[O \rho_{u,\xi}] \quad (5)$$

be a trajectory observable. There are three useful response objects.

First, the *trajectory-conditioned response* holds the complete record fixed:

$$\chi_{\xi}^{\text{cond}} = \partial_u O_{u,\xi} \big|_{u=0}. \quad (6)$$

This is a counterfactual branch derivative. It asks what the normalized state would do under a weak perturbation if precisely the same sequence of outcomes were imposed.

Second, the *fixed-record trajectory average* is

$$\chi^{\text{fix}} = \mathbb{E}_{p_0}[\chi_{\xi}^{\text{cond}}]. \quad (7)$$

It samples ordinary unperturbed records but does not differentiate their probabilities. We use “fixed-record” or “quenched conditional average” to remove the ambiguity in the informal phrase “trajectory-averaged response.”

Third, the *Born-ensemble response* differentiates the operational mean of the perturbed experiment:

$$\chi^{\text{B}} = \partial_u \sum_{\xi} p_u(\xi) O_{u,\xi} \bigg|_{u=0}. \quad (8)$$

Defining the path score

$$S_{\xi} = \partial_u \log p_u(\xi) \big|_{u=0}, \quad (9)$$

ordinary differentiation gives the exact identity

$$\chi^{\text{B}} = \mathbb{E}_{p_0}[\chi_{\xi}^{\text{cond}} + O_{0,\xi} S_{\xi}]. \quad (10)$$

Normalization implies $\mathbb{E}_{p_0}[S_{\xi}] = 0$, so a variance-reduced form is

$$\chi^{\text{B}} = \chi^{\text{fix}} + \text{Cov}_{p_0}(O_{0,\xi}, S_{\xi}). \quad (11)$$

The score is familiar from inference on continuously monitored systems [3], and recent work derives response bounds for quantum trajectory observables [4]. Equations (10) and (11) are nevertheless elementary likelihood-ratio identities: their role here is to identify the physical estimator unambiguously.

If separate experiments are run at $u = +\epsilon$ and $u = -\epsilon$, with records resampled from $p_{+\epsilon}$ and $p_{-\epsilon}$, the derivative of those two sample means estimates χ^{B} . By contrast, replaying the same baseline records at both signs estimates the two terms of Eq. (10) and exposes their decomposition.

TABLE II. Response taxonomy used throughout this note.

Object	Formula	Operational statement
Conditional response	$\chi_{\xi}^{\text{cond}} = \partial_u O_{u,\xi} _0$	Same complete record is imposed before and after the perturbation.
Fixed-record average	$\chi^{\text{fix}} = \mathbb{E}_{p_0}[\chi_{\xi}^{\text{cond}}]$	Average conditional sensitivity of ordinary unperturbed trajectories.
Score contribution	$\mathbb{E}_{p_0}[(O - \bar{O})S]$	Correction from the perturbation-induced redistribution of records.
Born response	$\partial_u \mathbb{E}_{p_u}[O_u] _0$	Repeated perturbed experiment with outcomes sampled from their perturbed Born probabilities.
Unconditional response	$\partial_u \text{Tr}(O\bar{\rho}_u) _0$	Identical to the Born response for a linear O , provided the complete instrument and classical feedback state are included.

C. Born ensemble versus unconditional state

The unconditional state is

$$\bar{\rho}_u = \sum_{\xi} p_u(\xi) \rho_{u,\xi}. \quad (12)$$

For any linear quantum observable,

$$\sum_{\xi} p_u(\xi) O_{u,\xi} = \text{Tr}(O\bar{\rho}_u). \quad (13)$$

Consequently the full Born density response can, in principle, be calculated from the differentiated unconditional instrument. Equation (10) is an unraveling of that same derivative. It is valuable because it separates what happens to a typical conditional state from what happens because the record distribution is reweighted.

This equivalence fails for nonlinear trajectory observables:

$$\mathbb{E}[S(\rho_{\xi})] \neq S(\bar{\rho}), \quad \mathbb{E}[|C_{ij,\xi}|^2] \neq |\bar{C}_{ij}|^2. \quad (14)$$

Entanglement, trajectory topology, purity, and a per-record fitted velocity must therefore be formed trajectory by trajectory before averaging. An adaptive schedule with finite classical memory may also require an enlarged classical-quantum channel rather than a system-only one; recent feedback full counting statistics makes this construction explicit [5].

III. EXACT GAUSSIAN TRAJECTORY RESPONSE

A. Covariance and sign convention

For the canonical engine we use

$$C_{ij} = \langle c_i^{\dagger} c_j \rangle, \quad G = 2C - \mathbf{1}, \quad (15)$$

so G is Hermitian with spectrum in $[-1, 1]$. There is a persistent transpose convention hazard in the wider repository: the alternative one-particle density matrix $\tilde{C}_{ij} = \langle c_j^{\dagger} c_i \rangle = C_{ji}$ transforms by the adjoint single-particle convention. All derivative signs must therefore be checked against the implemented matrix transformation, rather than inferred from a shorthand phrase such as $C = \langle c^{\dagger} c \rangle$.

The current H2 code implements the source as

$$G_{\epsilon} = D_{\epsilon} G D_{\epsilon}^{\dagger}, \quad D_{\epsilon} = e^{-i\epsilon P_0}, \quad (16)$$

and hence

$$X_0 \equiv \partial_{\epsilon} G_{\epsilon}|_0 = -i[P_0, G]. \quad (17)$$

Under Eq. (15), the many-body unitary producing Eq. (16) is $U_\epsilon = e^{+i\epsilon n_0}$. Writing instead the conventional pulse $e^{-i\epsilon n_0}$ replaces $\epsilon \mapsto -\epsilon$ and flips the overall response sign. It does not change the signed slope of the (k, ω) ridge or the reversal between wall orientations. Equation (16) is the authoritative convention for comparisons with the executable H2 product.

The impulse preserves total particle number and initially changes only coherences. Since all local number projectors commute, the instantaneous density response vanishes at the source time; the subsequent monitored dynamics converts the coherence tangent into a signed density pattern. This is fundamentally different from resetting a cell from empty to occupied, which injects a finite charge and is not a linear-response estimator.

B. One normalized measurement-and-reset event

Let $P = |\chi\rangle\langle\chi|$ be a normalized measured orbital, $Q = \mathbf{1} - P$, $m \in \{0, 1\}$ its outcome, and $\eta_m = 2m - 1$. Let $s \in \{0, 1\}$ denote the controller's target occupation. The Born probability is

$$p_m(G) = \frac{1 + \eta_m g}{2}, \quad g = \text{Tr}(PG) = \langle\chi|G|\chi\rangle. \quad (18)$$

For a nonzero-probability branch, projective measurement followed by perfect reset acts as

$$\boxed{\Phi_{m|s}(G) = Q(\mathbf{1} + \eta_m GP)^{-1}GQ + \eta_s P.} \quad (19)$$

The apparent inverse is only a rank-one resolvent. Define

$$L_m(G) = Q(\mathbf{1} + \eta_m GP)^{-1} = Q - \frac{\eta_m QGP}{1 + \eta_m g}. \quad (20)$$

At fixed P, m, s , exact differentiation gives the congruence

$$\boxed{D_G \Phi_{m|s}|_G[X] = L_m(G)XL_m(G)^\dagger.} \quad (21)$$

The target s does not enter the derivative because the reset covariance is a constant. Exact projectors make L_m singular: $L_m P = 0$. This rank loss is a physical part of the normalized response and must not be replaced by a pseudoinverse.

The current manuscript appendix uses the equivalent right-factor convention $J_m = L_m^\dagger$ and writes $D\Phi[X] = J_m^\dagger X J_m$. We use the left factor because chronological propagation then has its familiar order.

C. Chronological tangent and general parameter dependence

Along a fixed record, let L_e be Eq. (20) evaluated on the state immediately before event e . If the perturbation is only the initial impulse and does not change later projectors, then

$$X_{e+1} = L_e X_e L_e^\dagger, \quad (22)$$

$$X_D = A_{D:0} X_0 A_{D:0}^\dagger, \quad A_{D:0} = L_{D-1} \cdots L_1 L_0. \quad (23)$$

This is an exact derivative of the normalized conditioned trajectory, not a derivative of an unnormalized Kraus product and not a finite difference.

For a parameter that also changes the measured orbital, write $P = P(u)$, $Q = \mathbf{1} - P$, $A = \mathbf{1} + \eta_m GP$, and $X = \partial_u G$. Direct Fréchet differentiation of Eq. (19) gives

$$\begin{aligned} \partial_u \Phi_{m|s} = \dot{Q}A^{-1}GQ + Q \left[-A^{-1}\dot{A}A^{-1}G + A^{-1}\dot{X} \right] Q \\ + QA^{-1}G\dot{Q} + \eta_s \dot{P}, \quad \dot{A} = \eta_m (XP + G\dot{P}). \end{aligned} \quad (24)$$

Equation (24) is the appropriate starting point for a gauge-field or twist tangent, where the controller orbitals themselves carry Peierls phases. H2 needs only the simpler homogeneous recursion Eq. (22).

D. Exact chronological score

The record probability factorizes conditionally:

$$p_u(\boldsymbol{\xi}) = \prod_e p_{m_e}(G_e(u), P_e(u)). \quad (25)$$

The score is therefore accumulated event by event,

$$S_{e+1} = S_e + s_e, \quad S_0 = 0, \quad (26)$$

$$s_e = \frac{\eta_{m_e} \dot{g}_e}{1 + \eta_{m_e} g_e}, \quad (27)$$

$$\dot{g}_e = \text{Tr}(P_e X_e) + \text{Tr}(\dot{P}_e G_e). \quad (28)$$

For the H2 initial density impulse, $\dot{P}_e = 0$ after the kick. The tangent X_e contains the complete effect of all earlier nonlinear normalized updates. The score cannot be obtained by differentiating every probability at the initial covariance.

For a probe cell r with two-orbital projector P_r , the density and its conditional derivative are

$$n_{r,\boldsymbol{\xi}}(t) = \text{Tr}(P_r C_{\boldsymbol{\xi}}(t)), \quad (29)$$

$$\chi_{r,\boldsymbol{\xi}}^{\text{cond}}(t; t_0) = \frac{1}{2} \text{Tr}[P_r X_{\boldsymbol{\xi}}(t; t_0)]. \quad (30)$$

The full wall-density response is then

$$\begin{aligned} \chi_r^{\text{B}}(t; t_0) = \mathbb{E}_{p_0} \left[\frac{1}{2} \text{Tr}(P_r X_{\boldsymbol{\xi}}(t; t_0)) \right] \\ + \mathbb{E}_{p_0} \left[(n_{r,\boldsymbol{\xi}}(t) - \bar{n}_r(t)) S_{\boldsymbol{\xi}}(t; t_0) \right]. \end{aligned} \quad (31)$$

Equation (31) is the central H2 formula. Its first line is the fixed-record trajectory average; its second is the Born-score correction.

E. Finite differences as validation, not definition

The current production runner forms

$$\begin{aligned} \partial_{\epsilon} n_{\boldsymbol{\xi}} &\simeq \frac{n_{+\epsilon,\boldsymbol{\xi}} - n_{-\epsilon,\boldsymbol{\xi}}}{2\epsilon}, \\ S_{\boldsymbol{\xi}} &\simeq \frac{\log p_{+\epsilon}(\boldsymbol{\xi}) - \log p_{-\epsilon}(\boldsymbol{\xi})}{2\epsilon}, \end{aligned} \quad (32)$$

using common frozen schedules and outcomes. It uses $\epsilon = 10^{-3}$ and retains 5×10^{-4} and 2×10^{-3} as validation scales. This is a legitimate central-difference estimator and its GPU batching is efficient, but it is not the analytic tangent of Eqs. (22) and (26). The exact tangent removes step-size bias, cancellation error, and the need for two full replays per source. The central difference should remain as a small-system parity test over an ϵ convergence window.

IV. DENSITY RESPONSE AS A CHIRALITY DIAGNOSTIC

A. Response, correlation, and fluctuation–dissipation

Three density objects are often called a “density correlator” in informal discussion:

$$\begin{aligned} C_{nn}(r, t) &= \langle n(r, t) n(0, 0) \rangle_c, & \text{spontaneous correlation,} \\ \chi_{nn}^R(r, t) &= -i\theta(t) \langle [n(r, t), n(0, 0)] \rangle, & \text{closed equilibrium response,} \\ \chi_{nn}(r, t) &= \left. \frac{\delta \langle n(r, t) \rangle_V}{\delta V(0, 0)} \right|_{V=0}, & \text{operational susceptibility.} \end{aligned} \quad (33)$$

In a closed equilibrium system, Kubo theory relates the last two objects [6]. A monitored nonequilibrium trajectory generally does not obey the equilibrium fluctuation–dissipation theorem. The dissipator, normalized backaction, and record probabilities all respond. Open-system NESS response theory explicitly warns that the answer need not reduce to a simple system-only two-point correlator [7, 8]. For this project the safest definition is therefore the operational derivative in the last line of Eq. (33), evaluated exactly by Eq. (31).

Static density correlations can demonstrate a critical wall and determine a scaling exponent, but they do not determine the propagation direction. The retarded density response contains that missing arrow of motion.

B. Real-space pattern

Let x_w denote one domain wall and let $f_w(x)$ be a narrow, preregistered transverse window. For a source at (x_w, y_0) define

$$\chi_w(r, t) = \sum_x f_w(x) \chi_{(x, y_0+r), (x_w, y_0)}^B(t; t_0), \quad r = y - y_0 \pmod{N_y}. \quad (34)$$

A chiral response should satisfy the following conjunction:

1. the response remains concentrated in x near the chosen wall;
2. its signed or squared weight moves predominantly in one y direction;
3. the propagation direction reverses on the oppositely oriented wall;
4. a source in the bulk lacks the isolated directional wall feature;
5. a matched trivial or wall-off circuit lacks the same low-energy ridge;
6. the sign is stable against the source, late time origin, wall-window width, and fit window.

The density-phase impulse has zero net injected charge, so $\sum_r \chi_w(r, t)$ can be small or vanish and the response can be a signed dipole. A centroid must therefore use a declared nonnegative weight such as $|\chi|$ or $|\chi|^2$, while the signed response itself is retained for physics. A useful real-space directional statistic is

$$\mathcal{C}_w(t) = \frac{W_{w,+}(t) - W_{w,-}(t)}{W_{w,+}(t) + W_{w,-}(t)}, \quad (35)$$

where $W_{w,\pm}$ are response weights on the two directed arcs before wraparound.

C. Why a Fourier ridge is chiral

For a single low-energy chiral density branch, chiral Luttinger theory gives, up to normalization and Fourier conventions,

$$\chi_{nm}^R(k, \omega) \propto \frac{k}{\omega - v_w k + i0^+}. \quad (36)$$

This is the equilibrium reference prediction [9]. Density-coupled spectroscopy and real-time charge propagation have been used to expose chiral edge dispersions in closed lattice models [10, 11].

Monitoring replaces an infinitely sharp pole by a generally broadened mode,

$$\chi_w(k, \omega) \sim \frac{Z_w(k)}{\omega - v_w k + i\Gamma_w(k)} + \chi_w^{\text{reg}}(k, \omega), \quad (37)$$

or, for a stroboscopic channel eigenmode,

$$\lambda_w(k) = e^{-\gamma_w(k) + i\omega_w(k)}. \quad (38)$$

The peak location measures $\omega_w(k)$, while its linewidth reflects $\gamma_w(k)$. At low momentum one expects

$$\omega_w(k) \simeq v_w k, \quad v_{w_1} v_{w_2} < 0. \quad (39)$$

The density operator probes particle–hole density excitations, not the bare single-particle edge band. For a nearly linear edge they share the same characteristic velocity; band curvature, monitoring, and bulk leakage broaden the response into a continuum.

The lattice edge spectrum remains periodic over the Brillouin zone. Its topological feature is spectral flow connecting bulk continua, not literal nonperiodicity of an edge energy function. Equation (36) is a low-energy description near the crossing.

D. Discrete Fourier protocol and symmetry

After translating every source to $r = 0$ within its parent trajectory, use

$$\chi_w(k_j, \omega_\ell) = \sum_{t=0}^{T_R} \sum_{r=0}^{N_y-1} w_t e^{i\omega_\ell t - ik_j r} \chi_w(r, t), \quad (40)$$

with $k_j = 2\pi j/N_y$ and a declared causal time taper w_t . Ridge fitting uses the native frequency resolution; zero padding is a visualization choice. Because circuit time is discrete, quasifrequency is periodic modulo 2π per cycle, so the fit must remain inside a low-frequency window away from aliasing and short-time lattice structure.

For a real time-domain response,

$$\chi_w(-k, -\omega) = \chi_w(k, \omega)^*. \quad (41)$$

The unavoidable $(-k, -\omega)$ image is not a counterpropagating partner. One may restrict to $\omega > 0$ and compare the two momentum signs, or use a wedge asymmetry

$$\mathcal{A}_w = \frac{\sum_{(k,\omega) \in \mathcal{W}_+} W_w(k, \omega) - \sum_{(k,\omega) \in \mathcal{W}_-} W_w(k, \omega)}{\sum_{(k,\omega) \in \mathcal{W}_+} W_w(k, \omega) + \sum_{(k,\omega) \in \mathcal{W}_-} W_w(k, \omega)}, \quad (42)$$

where $\mathcal{W}_\pm$ are matched low-energy regions with $\text{sgn}(\omega k) = \pm 1$. W_w may be $|\chi_w|^2$ for robust ridge localization, but it must not be called the retarded spectral function. When phase information and conventions are controlled, $-\text{Im } \chi^R$ carries the sharper signed spectral meaning.

The two physical walls must never be summed before this analysis: their opposite velocities can cancel. Source positions and time origins are correlated variance-reduction samples inside a trajectory. They do not increase the number of statistically independent trajectories.

V. WHAT RECORD-WISE EVIDENCE CAN AND CANNOT PROVE

A. Distribution of conditioned responses

For every parent trajectory, H2 can report a record-conditioned real-space map, Fourier ridge velocity $v_{w,\xi}$, and asymmetry $\mathcal{A}_{w,\xi}$. The scientifically meaningful questions are then:

- Does the distribution concentrate at a nonzero sign?
- Does the sign reverse between the paired wall orientations?
- How large is the score correction relative to the fixed-record mean?
- Does the wrong-chirality tail shrink with N_y ?
- Are outliers associated with small Born denominators, leakage, or nonstationarity?

Finite Monte Carlo data cannot prove that *every* possible record is chiral. A complete record space is exponentially large and can include rare branches with anomalous response. The defensible trajectory-level statement is an almost-sure or high-probability claim supported by the distribution, quantiles, sign fraction, finite-size evolution, and preregistered failure counts. The Born mean can be chiral even if some conditioned records are not; conversely, a chiral fixed-record mean can be altered or canceled by the score term.

TABLE III. Three chirality clocks and their claim boundaries.

Route	Clock	Positive signature	Does not establish
H1 modular drift	Modular time generated by $K_A = -\log \rho_A$	Opposite packet drifts on opposite walls, formed trajectory by trajectory	Physical circuit-time transport or a Hall coefficient
H2 density response	Monitored circuit time	Wall-localized one-way $\chi(k, \omega)$ ridge; conditional and score pieces; orientation reversal	Quantized transport by itself
H3 frozen twist	Auxiliary twist angle ϕ	Continuous wall-resolved entanglement spectral flow for the same record	Born-ensemble frequency, typicality, or a measured DC current

B. H3 frozen-record flux flow is a different record-wise question

H3 freezes one complete physical record sampled at zero twist and replays that same record for $0 \leq \phi \leq 2\pi$. Conceptually one seeks a wall-resolved entanglement flow. The current executable restricts $C = (G + 1)/2$ to one half-torus containing all x and $0 \leq y < N_y/2$, selects the eigenmodes nearest occupation $1/2$, and assigns them to nominal wall windows by their x weight. Let $\nu_a(\phi)$ be the resulting overlap-tracked covariance eigenvalues and define the entanglement levels

$$\varepsilon_a(\phi) = \log \frac{1 - \nu_a(\phi)}{\nu_a(\phi)}. \quad (43)$$

Oriented crossings of $\varepsilon = 0$ diagnose branchwise spectral flow. The equilibrium analogy is Laughlin flux insertion on a cylinder [12]: a 2π twist shifts allowed k_y values by one momentum step. Bulk bands return to themselves as sets, while a chiral edge branch undergoes a level permutation through the gap. Oppositely oriented interfaces carry opposite permutations, so the total flow of the periodic two-wall sample vanishes.

The record must remain fixed because Berry holonomy or spectral flow requires a continuous family. Independently resampling $\xi(\phi)$ at every twist produces unrelated states, and no amount of eigenvalue sorting creates a physical eigenbundle. The family is defined only where its branch probability $p_\xi(\phi)$ remains nonzero and its relevant rank is constant.

The approved production H3 scope is one preregistered interface record and one matched-trivial record at 20×24 , each on a 65-point closed circle. The seam gauge places the nontrivial link between $y = 23$ and $y = 0$. H3 is therefore a representative branch-level diagnostic. It cannot establish the fraction of Born records with spectral flow, typicality, or ensemble quantization. H3 complements H2 but does not replace it.

VI. THREE CLOCKS FOR CHIRALITY

The project now has three primary chirality routes. Their agreement would be powerful precisely because their clocks and claims differ.

H1 asks whether the prepared state has a handed modular generator. H2 asks whether the physical monitored dynamics carries a causal density disturbance in one direction. H3 asks whether a continuous conditional branch has oriented flux spectral flow. Static entanglement scaling fixes critical content such as $c_L + c_R$, but it does not determine the propagation orientation $c_R - c_L$; at least one orientation-sensitive route is necessary.

VII. STEADY-STATE AND OPEN-SYSTEM RESPONSE

A. A steady state is not a response theory

It is tempting to “take the steady state and look at it.” A stationary covariance can reveal an occupation or purity gap and a topological marker, but it does not determine how the system responds. The same ρ_{ss} can be stationary under distinct generators with different relaxation poles, currents, and susceptibilities. One needs the steady state *and* the dynamical generator or quantum instrument, including how the perturbation couples to it. This distinction is central in classifications of quadratic Lindbladians, where steady-state topology and damping-spectrum topology are not interchangeable [13].

For a differentiable discrete unconditional channel

$$\rho_{n+1} = \mathcal{E}_u(\rho_n), \quad \mathcal{E}_0(\rho_{ss}) = \rho_{ss}, \quad (44)$$

the tangent obeys

$$X_{n+1} = \mathcal{E}_0(X_n) + (\partial_u \mathcal{E}_u)_0(\rho_{ss})u_n. \quad (45)$$

For harmonic $u_n = u_\omega z^n$, the stationary response contains the channel resolvent

$$X(\omega) = (z\mathbf{1} - \mathcal{E}_0)^{-1} (\partial_u \mathcal{E}_u)_0(\rho_{ss})u_\omega, \quad z = e^{-i\omega}. \quad (46)$$

The eigenvalues of $\mathcal{E}_0$ determine resonance positions and damping.

For a continuous Markovian generator,

$$\dot{\rho} = \mathcal{L}_u \rho, \quad (47)$$

one analogously obtains

$$\dot{X} = \mathcal{L}_0 X + (\partial_u \mathcal{L}_u)_0 \rho_{ss} u(t), \quad X(\omega) = (i\omega - \mathcal{L}_0)^{-1} (\partial_u \mathcal{L}_u)_0 \rho_{ss} u(\omega), \quad (48)$$

for the stated Fourier convention. Lindblad and NESS response theories are well established [7, 14–16]. Gaussian-channel response can also close at the covariance level [17]. These are unconditional theories; the exact trajectory tangent plus score is their record-resolved realization for the present instrument.

B. Deterministic mean-channel special case

The completed CPU mean-channel campaign studies

$$\dot{C} = -DC - CD + S, \quad DC_{ss} + C_{ss}D = S. \quad (49)$$

For a density-phase impulse with $\delta C(0) = -i[P_0, C_{ss}]$, its exact response is

$$\delta C(t) = -ie^{-Dt}[P_0, C_{ss}]e^{-Dt}. \quad (50)$$

This supplies a useful analytic calibration and a controlled special case of Eqs. (48) and (37). It contains no sampled measurement records, no nonlinear normalized branch, and no score term. It must not be presented as an H2 Born-trajectory result. Conversely, a gapped deterministic mean channel does not rule out slow or chiral conditioned trajectory structure.

VIII. NO-POSTSELECTION HALL TRANSPORT IN THE SPACE–TIME-DISORDERED CIRCUIT

A. Density chirality is already a coherent target

H2’s immediate claim is directionality: a localized, charge-preserving perturbation produces a wall-confined response whose dispersion has only one low-energy velocity sign, reversed on the other wall. This is sufficient to show a chiral domain-wall mode. It does not assign a quantized transport coefficient.

Open-system topology and transport are not generically locked. A topological steady covariance can coexist with nonquantized Hall response, and special conditions are required for quantization [16, 18]. Finite-range dissipative state preparation also obeys nontrivial locality constraints [19]. The manuscript should therefore keep “state topology,” “chiral density response,” and “quantized transport” as three separate claims.

B. Primary microscopic numerical definition

No field theory is needed to define or estimate the response. The basic data are only the covariance trajectory, the ordered measurement record, and the controller orbitals already used by the simulator.

Partition the periodic x direction into two complementary regions R_L and R_R , each containing one interface, with the two artificial partition cuts placed deep in the topological and trivial bulks. For the production $N_x = 20$ explicit-interface geometry, a convenient preregistered choice is

$$R_L : x = 0, \dots, 9, \quad R_R : x = 10, \dots, 19. \quad (51)$$

The physical interfaces lie on bonds (4,5) and (15,16), whereas the partition cuts lie near the midpoints of the topological and trivial sectors. For any one-particle region projector R_a , $a = L, R$, define the trajectory particle number

$$N_{a,\xi}(c) = \frac{1}{2} \text{Tr}[R_a(\mathbf{1} + G_\xi(c))]. \quad (52)$$

At event e , the normalized controller orbital at the instantaneous twist is $|\chi_e(\phi_e)\rangle$, with $P_e(\phi_e) = |\chi_e(\phi_e)\rangle\langle\chi_e(\phi_e)|$. If the Born outcome is $m_e \in \{0, 1\}$ and the perfect-correction target is $s_e \in \{0, 1\}$, the feedback ancilla changes the physical covariance by $(s_e - m_e)P_e$. Its contribution to region a is therefore the exactly known number

$$A_{a,\xi}(c) = \sum_{e \leq c} (s_e - m_e) \text{Tr}[R_a P_e(\phi_e)]. \quad (53)$$

The source-subtracted regional balance is

$$q_{a,\xi}(c) = N_{a,\xi}(c) - N_{a,\xi}(0) - A_{a,\xi}(c), \quad (54)$$

in units of electron number. Finally define the oriented reduced transverse transfer between the two walls by

$$q_{x,\xi}(c) = \frac{1}{2} [q_{R,\xi}(c) - q_{L,\xi}(c)]. \quad (55)$$

Because $R_L + R_R = \mathbf{1}$, exact perfect-correction arithmetic requires $q_L + q_R = 0$ up to numerical roundoff. Equation (55) consequently uses both regions as an internal continuity check rather than relying on an assumed current operator. Record by record, this reduced quantity also contains the measurement-innovation change of a conditioned regional expectation. It is therefore a precise and inexpensive monitored-channel pump estimator, but not yet a literal electrical current. This innovation cancels in the ordinary Born mean of a local compiled measurement; it matters critically for record-level current distributions.

Everything in Eqs. (52)–(55) is available microscopically. The cycle observer supplies $G(c)$; the record observer supplies $s_e - m_e$, the site, and the channel; and the twisted controller spinors supply

$$w_a(j, \nu; \phi) = \text{Tr}[R_a P_{j,\nu}(\phi)] = \sum_{i \in R_a} |\chi_{j,\nu,i}(\phi)|^2. \quad (56)$$

Thus the observable can be streamed as two scalar arrays per trajectory and cycle. No covariance history, Keldysh functional, or guessed bond-current operator is required.

C. Exact deterministic Born mean

For the canonical perfect-correction instrument, the Born mean of every linear covariance observable closes exactly. This removes even the need for trajectory Monte Carlo when the first target is the ensemble-mean Hall coefficient.

For one normalized rank-one mode P_e , let $Q_e = \mathbf{1} - P_e$ and let its target occupation be $s_e \in \{0, 1\}$. Averaging the two normalized measurement and reset branches with their actual Born probabilities gives

$$\overline{C}_{e+1} = Q_e(\phi_e) \overline{C}_e Q_e(\phi_e) + s_e P_e(\phi_e). \quad (57)$$

The nonlinear branch denominators cancel in the weighted sum. In the $G = 2C - \mathbf{1}$ convention this is

$$\overline{G}_{e+1} = Q_e \overline{G}_e Q_e + (2s_e - 1) P_e. \quad (58)$$

Equation (58) is already the public `mean_replacement=True` path of the canonical GPU engine. For a fixed exogenous site schedule it is the exact Born average over all intrinsic measurement words, not an approximation. Random schedules can be sampled in parallel; each GPU batch member then represents one schedule realization with its Born outcomes summed analytically.

The expected feedback source is equally microscopic. Immediately before event e ,

$$\bar{y}_e = \mathbb{E}[s_e - m_e] = s_e - \text{Tr}(P_e \bar{C}_e), \quad (59)$$

so that

$$\bar{A}_a(c) = \sum_{e \leq c} [s_e - \text{Tr}(P_e \bar{C}_e)] \text{Tr}(R_a P_e). \quad (60)$$

Propagating Eqs. (57) and (60) therefore gives the exact Born-mean version of Eqs. (54) and (55). The only sampling error is from an optional random schedule ensemble, not from Born outcomes.

An analytic microscopic tangent is also finite-dimensional. With dots denoting differentiation with respect to the applied flux,

$$\dot{\bar{C}}_{e+1} = \dot{Q}_e \bar{C}_e Q_e + Q_e \dot{\bar{C}}_e Q_e + Q_e \bar{C}_e \dot{Q}_e + s_e \dot{P}_e, \quad \dot{Q}_e = -\dot{P}_e. \quad (61)$$

This is a purely numerical matrix recursion through the microscopic event sequence. It computes the linear Born response without a finite difference, without a likelihood score, and without a correlation function. The score is needed only when one insists on resolving the derivative trajectory by trajectory before averaging; Eq. (57) has already performed that average exactly.

The most efficient campaign hierarchy is consequently:

1. propagate Eq. (57) under the flux protocol to establish whether the Born mean has a plateau;
2. validate its flux derivative either by Eq. (61) or paired finite differences;
3. only if the mean result is promising, run physical Born trajectories to measure variance, quantiles, and record concentration.

The current mean-replacement path needs one lightweight event observer to emit Eq. (59); this is smaller than implementing a new response formalism.

D. Microscopic electrical current from a local gate compilation

A literal current can also be calculated numerically, but the measurement must first be given a spatial implementation. For $n_{\text{shell}} = 1$, every controller orbital has support on at most 3×3 cells and two microscopic orbitals per cell. Choose a fixed nearest-neighbor complex-Givens tree $W_e(\phi)$ that rotates $|\chi_e(\phi)\rangle$ onto an anchor orbital at the controller center. One exact event is then compiled as

$$C_e \xrightarrow{W_e(\phi_e)} C_{e,1} \xrightarrow{\text{local number measurement and fSWAP reset}} C_{e,2} \xrightarrow{W_e(\phi_e)^\dagger} C_{e+1}. \quad (62)$$

The local readout and its collocated fresh ancilla exchange charge with the reservoir but do not define transverse motion. Spatial transport occurs during the explicitly routed charge-conserving Givens gates.

For every elementary one-particle gate u_g and half-system cut projector R , accumulate

$$\frac{q_{x,g}}{e} = \text{Tr}[R(u_g C_g u_g^\dagger - C_g)]. \quad (63)$$

Only gates whose declared spatial path crosses the cut contribute. The trajectory current is

$$Q_{x,\xi}^{\text{gate}} = \sum_{g \in \xi} q_{x,g}. \quad (64)$$

Unlike the reduced balance Eq. (55), Eq. (64) does not count a Bayesian measurement innovation as transported charge. The same gate contractions on the deterministic covariance Eq. (57) give the exact Born-mean electrical current.

The compilation is part of the microscopic model. Two local Givens trees can realize the same reduced rank-one update while distributing short-distance currents differently. A quantized large-scale coefficient should agree between reasonable local trees and under cut deformation. Required validation is eventwise equality with the current rank-one engine at zero field—identical outcome probability and final covariance to numerical tolerance—together with gate-resolved continuity on the enlarged system–ancilla circuit. The reduced source-subtracted pump is the inexpensive discovery calculation; the compiled gate current is the decisive calculation for the phrase “electrical Hall conductance” and for a strong record-concentration claim.

E. Immediately runnable sampled finite-step estimator

The current engine can already measure a microscopic linear impulse response without any dynamics modification. First draw a Born batch at zero twist and burn it in to $G_{0,\xi}$. From every same baseline covariance, launch two ordinary continuations with static seam twists $+\epsilon$ and $-\epsilon$. The outcomes in both continuations are newly Born sampled under their own twisted covariances; they must never be frozen to the zero-twist word.

A sudden twist step is an impulse of the circumferential electric field. With an orientation sign $\varsigma = \pm 1$ fixed once from the coordinate and flux conventions, the direct central estimator is

$$\hat{C}_{\text{step}}(c; \epsilon) = \varsigma \frac{\pi}{\epsilon} \left[\overline{q_x^{(+\epsilon)}(c)} - \overline{q_x^{(-\epsilon)}(c)} \right]. \quad (65)$$

The sign ς is not chosen after seeing the data. Reversing the twist must reverse the unscaled transfer, while reversing the programmed Chern orientation must reverse the inferred integer. A linear-response result requires an ϵ^2 extrapolation of the central estimate,

$$\hat{C}_{\text{step}}(\epsilon) = C_0 + b\epsilon^2 + O(\epsilon^4), \quad (66)$$

and a time window in which the integrated response has converged before finite size recurrence or uncontrolled late-time relaxation.

The executable sequence is:

1. run `run_markov_circuit` at $\phi = 0$ with `G_history=False`, `save=False`, and `return_data=True`; retain only `G_final`;
2. snapshot the GPU and NumPy random states immediately before the continuation;
3. build the $+\epsilon$ model, call `set_controller_twist`, restore the snapshot, and continue from `G_init=G_final` with random schedules and ordinary Born outcomes;
4. repeat from the identical baseline and random-state snapshot at $-\epsilon$;
5. use the cycle observer for Eq. (52) and the record observer with a precomputed table Eq. (56) for Eq. (53);
6. construct Eq. (65) separately for every independent parent trajectory, then average and bootstrap whole trajectory blocks.

Restoring the same random stream is a common-random-number variance reduction, not postselection. Each marginal run still compares its shared uniform random number with its own Born probability. Branches may and should diverge between $+\epsilon$ and $-\epsilon$. Perfect correction has a fixed number of random draws per event, making this coupling particularly clean.

A sensible first grid is

$$\epsilon \in \{\pi/8, \pi/16, \pi/32\}, \quad c \leq 2N_y, \quad (67)$$

rather than $\epsilon = 10^{-3}$. For $C = 1$, the raw one-sided signal is only $|q_x| \simeq \epsilon/(2\pi)$; an unnecessarily tiny step converts ordinary trajectory noise into an impossible sample requirement. The three finite steps allow the linear limit to be tested rather than assumed.

F. Direct quantized pump: the stronger numerical experiment

The clearest quantization test avoids dividing a small noisy signal by ϵ . After zero-twist burn-in, ramp the microscopic controller holonomy through one full circle,

$$\phi_c = 2\pi f(c/T_\phi), \quad f(u) = u - \frac{\sin(2\pi u)}{2\pi}, \quad c = 0, \dots, T_\phi, \quad (68)$$

where the smooth profile has zero endpoint velocity. At the beginning of each cycle, twist the controller orbitals to ϕ_c , visit a newly random site schedule, sample the actual Born outcomes, perform feedback, and update Eqs. (53)–(55). The provisional reduced-channel per-record pump estimator is

$$\hat{C}_{\text{pump},\xi} = \varsigma q_{x,\xi}(T_\phi), \quad (69)$$

because a 2π cycle transfers C particles in the ideal orientation. This is an order-one signal and is the recommended discovery observable. For the final electrical claim, replace q_x by $Q_{x,\xi}^{\text{gate}}/e$ from Eq. (64).

The current static-twist method already supplies all projector phases. A zero-engine-change pilot can run the entire ramp in one uninterrupted call to `run_markov_circuit`: the cycle observer records the state at cycle c and then calls `set_controller_twist(phi[c+1], gauge="seam")` so the next cycle uses the next absolute flux. The initial $c = 0$ observer call installs the first ramp value. This preserves the canonical cycle loop, newly random schedules, and live Born draws. It also avoids the initialization overhead and semantic ambiguity of repeated one-cycle runs.

For production provenance, the minimal canonical-engine extension is still a validated `controller_twist_schedule` argument, applied at the start of each cycle before its measurement updates. A seam-gauge ramp is the primary protocol. A time-dependent uniform-gauge comparison must not be advertised until the associated temporal gauge/frame kick has been implemented. In seam gauge, the weights $\text{Tr}(R_a P_e)$ are independent of ϕ because the twist changes component phases but not their magnitudes; they can be precomputed once. H3 cannot be repurposed: H3 rebuilds a static model at each flux and forces one zero-flux record, whereas Eq. (68) is one continuous physical trajectory with fresh Born outcomes.

Run both orientations of the flux cycle and form the flux-odd estimator

$$\hat{C}_{\text{pump}}^{\text{odd}}(T_\phi) = \frac{\zeta}{2} \left[\overline{q_x^{(+2\pi)}(T_\phi)} - \overline{q_x^{(-2\pi)}(T_\phi)} \right]. \quad (70)$$

It cancels flux-even stochastic drift and is the primary reduced-channel plateau estimator. The individual forward and reverse distributions remain mandatory outputs.

For each N_y and T_ϕ , report

$$\begin{aligned} \bar{C}_{\text{pump}} &= \frac{1}{S} \sum_{a=1}^S \hat{C}_{\text{pump},a}, \\ V_C &= \frac{1}{S-1} \sum_{a=1}^S (\hat{C}_{\text{pump},a} - \bar{C}_{\text{pump}})^2, \\ f_{\text{bad}}(\delta) &= \frac{1}{S} \sum_{a=1}^S \mathbf{1}(|\hat{C}_{\text{pump},a} - C_{\text{target}}| > \delta). \end{aligned} \quad (71)$$

The indicator in Eq. (71) is an ordinary classical indicator, not an identity operator. The Born sample mean tests a no-postselection plateau; decreasing V_C and f_{bad} with size and ramp time test record typicality.

G. Concrete pilot and production ladders

The first calculation is an exact Born-mean screen using `mean_replacement=True`, with a fixed random schedule per batch member. It requires only a lightweight event observer that emits Eq. (59); the flux itself can be advanced by the cycle callback just described. If this deterministic mean has no plateau, large trajectory sampling is not justified.

The first sampled no-engine-change pilot is:

- explicit interface, $N_x \times N_y = 20 \times 24$, $n_{\text{shell}} = 1$, perfect correction, random schedule;
- $S = 10$ independent baseline trajectories and $2N_y$ burn-in cycles;
- paired $\pm\epsilon$ continuations for $\epsilon = \pi/8, \pi/16$, each observed for $2N_y$ cycles;
- the matched-trivial controller and reversed twist as mandatory controls.

This costs one burn-in plus four continuations, approximately five ordinary $2N_y$ parent runs, but they remain sample-batched on the A100. It stores only regional charge, source, continuity residuals, and compact record diagnostics; no covariance histories are needed.

If the signal and signs pass, add $\epsilon = \pi/32$, use $S = 25$, and repeat at $N_y = 24, 32$ before the full size ladder. The ramp discovery run should start at 20×24 , $S = 5$, and $T_\phi = 4N_y, 8N_y$, followed by a pilot with $S = 10$ and $T_\phi = 4N_y, 8N_y, 16N_y$. Production uses $S = 25$ in shards of five, the interface forward ramp at all three rates, the interface reverse ramp at $8N_y$, and matched-trivial forward/reverse controls at $8N_y$. A 20×32 size check is promoted only after the measured timing and variance are acceptable. The required extrapolations are

$$T_\phi^{-1} \rightarrow 0, \quad N_y^{-1} \rightarrow 0, \quad S^{-1/2} \rightarrow 0 \text{ statistically}, \quad (72)$$

with burn-in convergence checked separately. A finite set of 25 records can support a plateau and an initial distributional statement, not a theorem about every record.

The numerical acceptance checks are:

1. $q_L + q_R = 0$ and $\Delta N_{\text{total}} = \sum_e (s_e - m_e)$ within floating-point tolerance;
2. odd response under flux reversal and Chern-orientation reversal;
3. a matched-trivial result converging to zero;
4. stability under moving both the twist seam and the two regional cuts;
5. agreement of seam and properly transformed uniform gauges;
6. convergence in step size or ramp rate, circumference, burn-in, and sample count;
7. retention of the complete per-record distribution rather than only its mean.

This microscopic program is sufficient by itself. The generating-functional language below is retained only as an optional analytic consistency check and is not part of the proposed numerical implementation.

H. Why the microscopic derivative is legitimate outside equilibrium

Linear response is a functional derivative around a reference dynamics, not an equilibrium privilege. Equilibrium supplies the Kubo–Martin–Schwinger condition, fluctuation–dissipation relations, and familiar spectral proofs of quantization. A causal response around a transient, periodic state, or nonequilibrium stationary path ensemble remains well defined without any of those simplifications [7, 15, 20].

For one explicit space–time realization the kernel

$$K_{xy}^R(n, n') = \left. \frac{\delta \langle I_x(n) \rangle}{\delta A_y(n')} \right|_{A=0}, \quad n > n', \quad (73)$$

depends on both times separately. After burn-in, stationarity of the Born path measure permits an ensemble Fourier transform in $n - n'$. Statistical stationarity is sufficient; an individual measurement word need not repeat. The random schedule and intrinsic Born outcomes therefore act as genuine exogenous and endogenous space–time disorder, respectively.

I. Optional generating-functional check

Let $K_{\xi}[A]$ be the complete history-dependent branch operator, including feedback conditioned on earlier entries of ξ . Couple the forward and backward branches to independent gauge fields A_+ and A_- , and insert a counting field χ_x into every elementary operation that transports charge through a declared x cut. The exact instrument generating functional is

$$\mathcal{Z}[A_+, A_-; \chi_x] = \sum_{\xi} q(\xi_{\text{ext}}) \text{Tr} [K_{\xi}[A_+; +\chi_x/2] \rho_{\text{ss}} K_{\xi}[A_-; -\chi_x/2]^{\dagger}], \quad (74)$$

where $q(\xi_{\text{ext}})$ is the field-independent probability of any exogenous schedule choices. The same classical record labels join the two contour branches, while all intrinsic outcomes are summed. Trace preservation gives the unitarity identity

$$\mathcal{Z}[A, A; 0] = 1. \quad (75)$$

Thus Eq. (74) is a literal discrete-time Schwinger–Keldysh construction for the adaptive instrument. It is not a postselected or representative-record object. Replica Keldysh formulations for monitored jump processes and hybrid classical–quantum full counting statistics provide closely related precedents [5, 21].

Define

$$A_c = \frac{A_+ + A_-}{2}, \quad A_q = A_+ - A_-, \quad W = -i \log \mathcal{Z}. \quad (76)$$

Up to the declared sign and charge normalization, the physical current and retarded response are

$$\bar{j}_x(n) = \frac{\delta W}{\delta A_{x,q}(n)} \Big|_{A_q=0}, \quad (77)$$

$$K_{xy}^R(n, n') = \frac{\delta^2 W}{\delta A_{x,q}(n) \delta A_{y,c}(n')} \Big|_{A_q=0}. \quad (78)$$

If A_y enters only through a Hamiltonian term $-A_y J_y$, this reduces to a retarded current–current correlator plus the diamagnetic/contact term. In the present circuit the field also changes controller orbitals, projectors, Kraus maps, and feedback operations. The exact insertion is therefore an instrument vertex $\partial_{A_y} \mathcal{E}$, not generically a second current operator. A symmetrized current autocorrelation measures noise; outside equilibrium no fluctuation–dissipation theorem converts it into Eq. (78).

J. The Born score inside the Keldysh functional

At equal physical fields,

$$\bar{I}_x[A] = \sum_{\xi} p_A(\xi) I_{x,\xi}[A]. \quad (79)$$

Differentiating gives

$$\frac{\delta \bar{I}_x}{\delta A_y} \Big|_0 = \mathbb{E}_0[K_{xy,\xi}^{\text{cond}}] + \mathbb{E}_0[(I_{x,\xi} - \bar{I}_x) S_{y,\xi}], \quad (80)$$

with

$$K_{xy,\xi}^{\text{cond}} = D_G I_x[X_y] + \partial_{A_y} I_x, \quad (81)$$

$$S_{y,\xi} = \delta_{A_y} \log p_A(\xi) \Big|_0. \quad (82)$$

The gauge tangent has the inhomogeneous recursion

$$X_{y,e+1} = D_G \Phi_e[X_{y,e}] + \partial_{A_y} \Phi_e. \quad (83)$$

When the controller projector also depends on A_y , both the map tangent and the score contain the corresponding $\dot{P}$ terms. Equation (80) is exactly what the doubled functional produces: the first term moves the normalized state along a fixed word, while the second differentiates that word’s Born weight. The score is therefore not an optional correction. It is the classical record-sector part of the physical no-postselection response.

For Monte Carlo analysis one can retain the single-record influence estimator

$$\hat{K}_{xy,\xi} = K_{xy,\xi}^{\text{cond}} + (I_{x,\xi} - \bar{I}_x) S_{y,\xi}, \quad \mathbb{E}[\hat{K}_{xy,\xi}] = K_{xy}^{\text{B}}. \quad (84)$$

Its distribution is meaningful for diagnosing typicality, but its score term is an influence-function contribution, not the response of a postselected branch.

K. The model-specific conservation obstruction

The reduced physical fermion system is not charge conserving. When a measurement outcome disagrees with its controller target, perfect correction uses a fresh filled or empty ancilla and an fSWAP. Total $U(1)$ charge is conserved only in the enlarged system–ancilla implementation. The physical system obeys a sourced circuit-time continuity equation,

$$\Delta n_{\mathbf{r}} + (\nabla_{\text{lat}} \cdot j)_{\mathbf{r}} = s_{\mathbf{r}}^{\text{anc}}, \quad (85)$$

not a source-free one. Consequently the current reduced covariance engine and its controller twist do not, by themselves, define a conventional electromagnetic Hall current.

There are two honest observables:

1. *Electrical Hall response of a specified dilation.* Assign physical paths to every basis-rotation gate and ancilla port, Peierls-dress every charge-moving gate and fSWAP, and insert the counting field across all paths crossing the cut. Equation (85) then becomes source free on the enlarged space.
2. *Generalized monitored-circuit pump.* Use the controller holonomy as a parameter and explicitly subtract the known ancilla source. This is immediately useful, but should not be called electrical Hall conductance until its electromagnetic dilation is fixed.

This distinction is physical: inequivalent local Stinespring compilations can realize the same reduced Kraus map while routing microscopic charge along different paths. An electrical current is therefore not an intrinsic property of the reduced instrument alone. Independence of the quantized coefficient from reasonable local compilations would itself be a nontrivial robustness test.

At perfect correction the source subtraction is especially simple. If the measured occupation is $m \in \{0, 1\}$ and the reset target is $s \in \{0, 1\}$, then

$$C_{\text{after}} - C_{\text{meas}} = (s - m)P. \quad (86)$$

For a one-particle region projector R , the exact injected charge is

$$S_{R,e}^{\text{anc}} = e(s_e - m_e) \text{Tr}(RP_e). \quad (87)$$

The executable already records $s_e - m_e$ as the event transfer. With inward current positive, a practical source-subtracted regional transfer is

$$Q_{\partial R}^{\text{sub}} = e \Delta N_R - \sum_{e \in \text{ramp}} S_{R,e}^{\text{anc}}. \quad (88)$$

Equation (87) is exact for perfect correction. If feedback can fail, the code must archive the actual reset transfer rather than merely the requested target minus outcome.

L. What quantization would mean in Keldysh language

In units $e = \hbar = 1$, the desired parity-odd long-distance functional is

$$\begin{aligned} W_{\text{top}} &= \frac{C}{4\pi} \int d^3x \epsilon^{\mu\nu\lambda} (A_{\mu,q} \partial_\nu A_{\lambda,c} + A_{\mu,c} \partial_\nu A_{\lambda,q}) \\ &= \frac{C}{2\pi} \int d^3x \epsilon^{\mu\nu\lambda} A_{\mu,q} \partial_\nu A_{\lambda,c} + \text{boundary term}. \end{aligned} \quad (89)$$

It yields

$$j_x = \frac{C}{2\pi} E_y, \quad \sigma_{xy} = C \frac{e^2}{h}. \quad (90)$$

Additional causal dissipative kernels and terms quadratic in A_q encode relaxation and noise. They do not by themselves forbid the topological term. Keldysh Chern–Simons response has been established for specially structured number-conserving open systems with a pure stationary state [22]. That result is a template, not a theorem for the present circuit: pure conditioned trajectories generally average to a mixed Born state unless all positive-probability branches lie on the same ray.

For this adaptive process integer C is therefore a conjectural response coefficient that must be demonstrated under explicit hypotheses:

1. exact $U(1)$ gauge covariance of the full local dilation;
2. locality or controlled quasilocality of all charge-moving operations;
3. a stationary, ergodic late-time Born process;
4. a bulk dynamical, mobility, or rapid-mixing gap that remains controlled throughout the twist cycle;
5. vanishing longitudinal DC response in the scaling limit;

6. large-gauge closure and invariance under seam and cut deformation;
7. a stated order of transient, field, slow-ramp, thermodynamic, and statistical limits.

Static disorder can coexist with almost-sure quantized Hall response under mobility-gap hypotheses [23, 24]. Quantized Floquet pumping can survive spatial disorder and, over a controlled regime, temporal noise [25, 26]. These are important precedents, but their disorder is not the endogenous, state-dependent Born record of an adaptive instrument.

The caution is equally important. A conserved state Chern number after a quench need not imply quantized Hall response, and dissipatively prepared topology and transport can separate [18, 27]. Thus the proposed experiment would not merely remeasure the known trajectory invariant.

M. Three meanings of record-independent quantization

The following claims have strictly increasing strength:

1. *Born-mean plateau:*

$$\frac{\mathbb{E}_B[Q_{x,\xi}]}{e} \longrightarrow C. \quad (91)$$

This is already a physical no-postselection result, but it can hide a broad or bimodal record distribution.

2. *Born-typical concentration:*

$$\Pr_B(|Q_{x,\xi}/e - C| > \delta) \longrightarrow 0 \quad \text{for every fixed } \delta > 0. \quad (92)$$

This is the natural analogue of an almost-sure disorder statement and the strongest realistic meaning of “record independent.”

3. *Uniform every-word quantization:* every allowed finite record, including arbitrarily rare words, gives exactly the same integer. This would require a uniform gap and is neither needed nor currently plausible.

A quantized linear-response coefficient does not logically require the entire finite-time counting distribution to collapse to a delta function; irreducible shot noise can remain. The full counting statistics should therefore report the mean response, response variance, higher cumulants, quantiles, and bad-record fraction separately. Exact claims concern specified asymptotic limits, never a finite sample of 25 trajectories.

N. Dynamical bulk–boundary correspondence

Suppose two bulk regions have Born Keldysh coefficients C_L and C_R . At their interface, the gauge variation of Eq. (89) has an uncanceled boundary term proportional to

$$\Delta C = C_L - C_R. \quad (93)$$

Gauge invariance of the full dilation then requires a wall anomaly with net chirality ΔC . This is the anomaly-inflow statement of bulk–boundary correspondence, formulated for the Born-summed dynamical functional rather than a ground-state Hamiltonian.

H2 tests one side of this relation: wall-localized, signed propagation. It does not measure the coefficient C . H3 tests branchwise auxiliary entanglement spectral flow under a frozen record. It is not a physical flux ramp and does not sum the flux-dependent Born process. A quantized bulk pump or Kubo coefficient agreeing with the H2 wall chirality would supply the missing side. If Eq. (92) also holds, the result would be genuinely stronger: dynamical bulk–boundary correspondence would survive endogenous space–time measurement disorder without postselection.

O. Two operational protocols

1. *Protocol A: full-dilation retarded Hall kernel*

Use a homogeneous topological torus or cylinder first, rather than the two-wall geometry. Burn in to the stationary Born process, apply a weak time-dependent A_y to every affected operation of the local dilation, and count the full

charge crossing an x cut. Evaluate Eq. (78) either from the exact inhomogeneous state tangent plus score or from common-schedule $\pm A$ replays as a validation. After testing stationarity, extract the antisymmetric response, take the thermodynamic or mobility limit before the DC limit, and verify $\sigma_{xx} \rightarrow 0$.

For a stationary augmented channel $\mathbb{E}_0$, the discrete causal kernel has the transparent form

$$K_{O,A}^R(\ell) = \Theta(\ell) \langle\langle O | \mathbb{E}_0^{\ell-1}(\partial_A \mathbb{E})_0 | \varrho_{ss} \rangle\rangle + K_{\text{contact}}. \quad (94)$$

Classical feedback memory can be included in the augmented state. This is the discrete NESS analogue of a Lindblad resolvent, not an equilibrium fluctuation formula.

2. Protocol B: no-postselection Laughlin pump

The numerically cleaner quantization test is a physical y -flux ramp

$$\phi_y : 0 \rightarrow 2\pi, \quad A_y = \phi_y/N_y. \quad (95)$$

Outcomes must be newly sampled from the instantaneous flux-dependent Born process; freezing them would convert the experiment back into H3. Count the full-dilation charge, or provisionally the source-subtracted charge Eq. (88), crossing an x cut. With orientation sign fixed in advance, the target is

$$Q_x/e \rightarrow \pm C. \quad (96)$$

The associated full counting statistics can be generated by the tilted instrument

$$\mathcal{E}_{A,\chi}(\rho) = \sum_m K_m[A; +\chi/2] \rho K_m[A; -\chi/2]^\dagger, \quad (97)$$

with

$$\begin{aligned} \langle Q_x \rangle &= \partial_{i\chi} \log \mathcal{Z}|_{\chi=0}, \\ \partial_A \langle Q_x \rangle &= \partial_A \partial_{i\chi} \log \mathcal{Z}|_{A=\chi=0}. \end{aligned} \quad (98)$$

Tilted generators and trajectory large deviations are established tools [28, 29].

A discovery pilot can use the current 20×24 or 20×32 explicit interface, $n_{\text{shell}} = 1$, perfect correction, random schedules, five to ten Born trajectories, and ramp times $T_\phi = 4N_y, 8N_y, 16N_y$ after a burn-in of order $2N_y$. It should stream wall/strip charge, total system charge, and spatially resolved ancilla source rather than covariance histories. This pilot can establish feasibility and sign conventions; it cannot establish a quantized thermodynamic coefficient.

P. Controls and implementation consequences

The minimum controls are:

1. trivial controller, reversed Chern orientation, and reversed ramp;
2. seam versus uniform gauge, seam translation, and cut deformation;
3. a discrete Ward identity including the ancilla source, together with Eq. (75) and $\mathbb{E}[S] = 0$;
4. random versus raster schedules and controlled measurement noise;
5. probe-amplitude, frequency or ramp-rate, size, and wall-separation scaling;
6. longitudinal response and full counting noise;
7. at least two local physical compilations of the dilation if the result is called electrical Hall conductance.

The existing static method `set_controller_twist` is useful but insufficient. A new versioned campaign would need a time-dependent twist schedule inside the canonical GPU engine, ordinary Born sampling at each instantaneous twist, a cut/source observer, and the $\dot{P}$ terms in the gauge tangent and score. A time-dependent uniform gauge also requires the proper inter-cycle gauge or scalar-potential transformation; repeatedly rebuilding static H3 models is not a physical ramp. The implementation should validate eventwise gauge derivatives against $\pm\delta\phi$, 2π closure, seam/uniform equivalence, source-subtracted continuity, and exact small-system branch enumeration before production.

Quantized response need not always be framed as conventional DC transport. Frequency-integrated circular dichroism and certain pulse-to-density protocols can also tie responses to topological invariants in closed systems [30, 31]. Those alternatives do not make the H2 density ridge quantized, nor do they remove the dilation issue for electrical transport.

TABLE IV. Literature map and project-specific distinction.

Established framework	What is established	What remains specific here
Closed Kubo and chiral edge theory [6, 9] Lindblad/NESS response [7, 8, 14–16]	Retarded density response and a one-way pole $\omega = vk$ for a chiral edge. Generator linearization, resolvents, geometric response, and conditions under which open Hall response can remain quantized.	Adaptive normalized backaction, record dependence, and the Born score. Discrete rank-losing record-conditioned Gaussian instrument and branch distribution.
Keldysh topology and monitored field theory [20–22]	Nonequilibrium response functionals, Chern–Simons response in structured open systems, and Keldysh representations of monitored processes.	A doubled, history-dependent feedback instrument whose Born score and physical dilation are kept explicit.
Gaussian process response [17]	Covariance-level response of Gaussian channels.	Fermionic projective normalized map with exact outcome conditioning and feedback.
Quantum trajectory likelihood and response [3, 4]	Record likelihoods, tangent inference, Fisher information, and bounds on trajectory-observable response.	Local density impulse, exact branch congruence, and score-completed wall susceptibility in this circuit.
Trajectory large deviations [28]	Tilted generators and statistics of record observables.	Spatially resolved class-A domain-wall response and its conditional/Born comparison.
Feedback and monitored topology [32–34]	Topological quantum channels, monitored Lyapunov spectra, and boundary modes.	Density-response spectroscopy of finite-range adaptive Gaussian Born trajectories, including rank loss and score reweighting.
Dissipative topology [18, 19, 35]	Constraints on dissipative preparation and examples of trajectory/open-system topology.	Direct comparison among fixed-record, Born, mean-channel, and frozen-holonomy diagnostics on the same programmed wall.
Disordered topological transport [23–26]	Almost-sure static-disorder quantization and dynamical pumping robust to spatial or temporal disorder under controlled hypotheses.	Endogenous Born space–time disorder, score-completed response, and record-concentration tests in an adaptive circuit.

IX. RELATION TO THE LITERATURE AND NOVELTY BOUNDARY

Table IV positions the construction without claiming that monitored-system response theory itself is new. The plausible project-specific contribution is therefore a synthesis:

An exact, record-conditioned tangent response for a normalized, rank-losing adaptive fermionic Gaussian instrument, completed by the many-body Born likelihood score and analyzed as a wall-resolved momentum–frequency density susceptibility across a topological trajectory ensemble.

The more ambitious follow-on contribution would be a separate statement:

A gauge-covariant, no-postselection adaptive fermion instrument has a Born Hall coefficient locked to an integer, with its response distribution concentrating over endogenous space–time measurement records and matching the chiral wall anomaly.

Nothing in the present trajectory Chern data, H2 ridge, or H3 frozen holonomy already proves this second statement. The individual ingredients are not new. A formal novelty claim should use “to our knowledge” only after a broader systematic search. In particular, recent topology of discrete quantum feedback [32] is a close neighboring field that must be discussed, while monitored free-fermion classifications [34] supply the appropriate tangent/Lyapunov comparison.

X. CURRENT IMPLEMENTATION AND EXACT-TANGENT UPGRADE

A. Current production contract

The current H2 production choices are listed in Table V. They are taken from the audited working campaign and executable configuration, not from the superseded numerical-campaign copy inside the draft directory.

The H2 runner already preserves the essential estimator order: it freezes each baseline schedule and outcome record, central differences the conditional density and its chronological log likelihood, multiplies the latter by the baseline

TABLE V. Current H2/H3 contract as of August 18, 2026.

Quantity	Current choice
H2 geometry	$N_x = 20$; $\alpha_{\text{top}} = 1$, $\alpha_{\text{triv}} = 30$; $n_{\text{shell}} = 1$; topological columns $x = 5, \dots, 15$; interfaces on bonds (4, 5) and (15, 16); periodic y
H2 circumference	$N_y = 32, 48, 64$
Independent trajectories	$S = 25$, in five shards of five
Schedule law	Random serial only
Physical duration	$2N_y$ cycles from independent product-state starts
Sources	Eight translated y_0 positions on each of two wall orientations
Time origins	Four late origins per parent trajectory
Response horizon	$N_y/2$
Wall window	Half-width one cell in current executable config
Saved decomposition	Conditional derivative, likelihood-score contribution, total Born response, and within-trajectory source variance. The cumulative scalar score itself is not currently archived.
Current derivative	Central $\pm\epsilon$ replay with $\epsilon = 10^{-3}$; validation at 5×10^{-4} and 2×10^{-3}
Statistics	Whole parent trajectory is the bootstrap unit
H3	One interface and one matched-trivial record at 20×24 ; 65 twist points per closed circle; seam $y = 23 \rightarrow 0$; half-torus entanglement spectrum with wall-weight classification

density, translates sources within each trajectory, and retains the trajectory axis for later bootstrapping. It does not yet propagate Eqs. (22)–(26). The repository also does not yet contain the final downstream H2 Fourier/ridge merger; the current compact product is the real-space input to that analysis. A separate geometry caveat is also recorded here: the canonical CPU class currently uses an $N_x/3$ default wall width, whereas the production GPU H2/H3 geometry uses $N_x/4$. Production-response claims must use the GPU geometry stated in Table V, or explicitly reconcile the defaults before combining CPU and GPU results.

B. Recommended exact-tangent implementation

The upgrade should preserve the immutable finite-difference campaign rather than silently changing its schema. A new versioned H2 product should:

1. initialize a batched tangent $X_0 = -i[P_0, G]$ for every live source;
2. propagate the base covariance once through the canonical GPU engine;
3. update all source tangents with the exact local or dense rank-one Jacobian inside the canonical GPU class, respecting repository policy;
4. accumulate the scalar score increment Eq. (27) on the same device;
5. stream only $\text{Tr}(P_r X)/2$, the baseline density, score, wall leakage, denominator minima, and compact diagnostics;
6. archive an explicit schema/version and the exact engine hash;
7. compare against the existing $\pm\epsilon$ implementation without overwriting either result.

This removes two full continuations per source and should materially reduce H2 runtime, although the realized speedup depends on how many source tangents fit in the A100 batch and must be benchmarked rather than assumed.

C. Validation gates

Before the analytic tangent replaces the finite difference as the primary estimator, require:

1. event-level agreement of Eq. (21) with central differences for occupied and empty outcomes, mixed and nearly pure states, local and dense backends, and gain and loss targets;
2. complete frozen-record agreement over a convergent ϵ window;
3. agreement of the accumulated score with central differences of the complete chronological log probability;

4. exact small-system branch enumeration of Eq. (10);
5. $\mathbb{E}[S] = 0$ within whole-trajectory uncertainty, with the centered score estimator used in production analysis;
6. CPU/GPU parity for identical frozen words;
7. explicit counts of zero-probability/censored branches and minimum Born denominators;
8. sign-convention parity with Eq. (16).

XI. ANALYSIS AND CLAIM PROTOCOL

A. Per-trajectory construction before averaging

For each independent parent trajectory and each wall:

1. average translated sources and permitted late origins only as correlated variance reduction inside that trajectory;
2. retain $\chi_{w,\xi}^{\text{cond}}(r, t)$, $\chi_{w,\xi}^{\text{score}}(r, t)$, and their sum;
3. record transverse wall retention and bulk leakage;
4. Fourier transform each component with identical conventions;
5. extract ridge slope, linewidth, wedge asymmetry, and fit quality;
6. only then form ensemble means and bootstrap intervals over complete parent trajectories.

Spatial sites, sources, time origins, and the two walls are not independent Monte Carlo samples.

B. Primary figures

A compact but complete H2 presentation contains:

1. real-space $\chi_w(r, t)$ heatmaps for both wall orientations;
2. the full Born $\chi_w(k, \omega)$ for both walls, with common axes;
3. an inset decomposing conditional and score contributions;
4. the distribution of $v_{w,\xi}$ or $\mathcal{A}_{w,\xi}$;
5. a matched trivial or bulk-source control;
6. a size panel for signed velocity, wrong-chirality weight, and wall retention.

C. Acceptance and downgrade language

The strong H2 statement requires a low-energy ridge whose velocity sign is stable, reverses between the paired walls, remains wall localized, survives whole-trajectory uncertainty and size scaling, and is absent in the matched control. Both terms in Eq. (31) belong in the headline Born response.

If only χ^{fix} is chiral, report a chiral fixed-record conditional response and state that the physical Born response is unresolved or different. If only the mean is chiral while per-record signs are broad, do not claim that typical individual trajectories share the same response. If real-space drift is visible but the Fourier ridge is unresolved, report directional spreading, not a measured dispersion. If a ridge exists without wall reversal or a control null, report schedule-induced directional response rather than topological chirality. At no point should the H2 ridge be called a quantized Hall conductance.

XII. COHERENT RESEARCH QUESTIONS EXPOSED BY THE FRAMEWORK

The discussion suggests several questions that are logically independent and could support a later paper beyond the immediate H2 analysis.

A. Conditional versus Born chirality

How large is

$$\Delta\chi = \chi^{\text{B}} - \chi^{\text{fix}} = \text{Cov}(n, S), \quad (99)$$

and can the score change the direction, velocity, or only the spectral weight? The answer measures how strongly the local perturbation reshapes the record distribution rather than merely changing conditional states.

B. Self-averaging and trajectory typicality

Does the distribution of $v_{w,\xi}$ narrow with N_y and observation time, or does a broad record-dependent chiral response survive? A narrow distribution would connect the Born response to an almost-sure trajectory statement; a broad distribution would make the mean insufficient as a description of typical monitored runs.

C. Response poles versus tangent Lyapunov modes

Does the linewidth $\gamma_w(k)$ extracted from $\chi(k, \omega)$ agree with decay rates in the wall-localized tangent cocycle? The two use the same normalized derivative but different source and projection protocols. Agreement would connect observed chiral propagation to the slow mechanism responsible for purification; disagreement would identify distinct sectors.

D. Trajectory, mean-channel, and feedback-channel topology

Which wall features survive after discarding the record? Monitored Lyapunov topology, dissipative topology, and discrete feedback-channel topology are active neighboring subjects [32–35]. The present circuit can compare all three descriptions on the same microscopic controller instead of assuming their equivalence.

E. Record Fisher information and response cost

The same derivatives entering the score determine the Fisher information carried by the record. One may ask whether a robust wall response requires the record to learn the injected perturbation, and whether trajectory-response bounds [4] become tight near the wall. This is a question about the classical information channel of monitoring, not a substitute for the density susceptibility.

F. Gauge response and quantized pumping

Can a gauge-covariant extension of the adaptive instrument support a robust integer Born pump, and how is it related to the representative fixed-record H3 flow? This requires the full program of Sec. VIII. It is a plausible separate work, not a prerequisite for demonstrating chirality in the current manuscript.

XIII. RECOMMENDED PAPER-LEVEL STATEMENT

If the planned data pass their controls, the clean central statement is:

A local number-phase impulse produces a wall-localized density response with a single low-energy propagation direction. The momentum–frequency ridge reverses between the two oppositely oriented domain walls. This directionality is visible in the distribution of fixed-record trajectory responses and survives the Born likelihood-score completion required for the physical ensemble.

This statement is stronger than static criticality and more operational than modular-time motion, while remaining narrower and safer than a quantized transport claim. H1 modular drift and H3 frozen-record spectral flow can be presented as complementary state-level and branch-topology evidence. The conditional, fixed-record averaged, Born, and unconditional objects should be named explicitly every time; “trajectory average” without a definition is no longer adequate.

Appendix A: Derivation of the fixed-projector covariance congruence

Let

$$A = \mathbf{1} + \eta GP, \quad \Phi(G) = QA^{-1}GQ + \eta_s P. \quad (\text{A1})$$

At fixed P , $\delta A = \eta XP$ and

$$\begin{aligned} D\Phi[X] &= Q \left[-A^{-1}(\eta XP)A^{-1}G + A^{-1}X \right] Q \\ &= QA^{-1}X \left[\mathbf{1} - \eta PA^{-1}G \right] Q. \end{aligned} \quad (\text{A2})$$

The rank-one identities

$$\begin{aligned} QA^{-1} &= Q - \frac{\eta QGP}{1 + \eta g} = L, \\ \left[\mathbf{1} - \eta PA^{-1}G \right] Q &= Q - \frac{\eta PGQ}{1 + \eta \bar{g}} = L^\dagger \end{aligned} \quad (\text{A3})$$

give

$$D\Phi[X] = LXL^\dagger, \quad (\text{A4})$$

which proves Eq. (21). The derivation assumes $1 + \eta g > 0$, exactly the condition that the selected branch has nonzero Born probability.

Appendix B: Monte Carlo form of the Born identity

For S independent baseline trajectories $\xi_a \sim p_0$, the direct unbiased estimator is

$$\hat{\chi}_{\text{direct}}^{\text{B}} = \frac{1}{S} \sum_{a=1}^S [\chi_{\xi_a}^{\text{cond}} + O_{\xi_a} S_{\xi_a}]. \quad (\text{B1})$$

Because $\mathbb{E}[S] = 0$, an exactly unbiased, sample-centered form uses the ordinary sample covariance,

$$\hat{\chi}_{\text{cov}}^{\text{B}} = \overline{\chi^{\text{cond}}} + \frac{1}{S-1} \sum_{a=1}^S (O_{\xi_a} - \bar{O})(S_{\xi_a} - \bar{S}). \quad (\text{B2})$$

Centering only O but retaining the prefactor $1/S$ introduces an $O(1/S)$ bias; Eq. (B2), or an equivalent leave-one-out construction, removes it. The direct estimator Eq. (B1) can have larger variance because a finite sample generally has $\bar{S} \neq 0$.

Sources and late origins inside trajectory a should first be combined into a single vector-valued observation Y_a . Bootstrap resampling acts on the S complete Y_a blocks, preserving all within-trajectory correlations and the paired two-wall structure.

Appendix C: Checklist for reading an H2 archive

Before interpreting a saved response product, verify:

1. engine hash, covariance convention, source-kick sign, and schema;
2. physical geometry, explicit wall bonds, wall windows, and orientation labels;
3. independent trajectory count and shard/sample indices;
4. initialization, total duration, origin cycles, and response horizon;
5. random schedule law and whether schedules/outcomes were frozen in the derivative;
6. derivative method: finite difference or exact tangent;
7. finite-difference ϵ audit, if applicable;
8. conditional, score, and total-response arrays separately;
9. minimum Born denominator, replay residual, and censored branch counts;
10. wall retention, bulk leakage, and total-charge drift;
11. native Fourier resolution, time window, zero padding, and wedge definition;
12. whole-trajectory bootstrap and paired-wall reversal test.

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